Imperial College
London

Lecture 6
Instruction Set Architecture
(RISC-V ISA)

Peter Cheung
Imperial College London

URL: https://elec50010-iac.github.io/course-website/
email: p.cheung@imperial.ac.uk

RISC-V

- Developed by Krste Asanovic, David Patterson and colleagues at UC Berkeley in 2010
- First widely accepted **open-source** computer architecture
- Underlying design principles:
  1. **Simplicity favours regularity**
  2. **Make the common case fast**
  3. **Smaller is faster**
  4. **Good design demands good compromises**

Design Principles

Principle 1: Simplicity favors regularity

- Consistent instruction format
- Same number of operands (two sources and one destination)
- Easier to encode and handle in hardware

Principle 2: Make the common case fast

- RISC-V includes only simple, commonly used instructions
- Hardware to decode and execute instructions can be simple, small, and fast
- More complex instructions (that are less common) performed using multiple simple instructions
- RISC-V is a ***reduced instruction set computer*** **(RISC)**, with a small number of simple instructions
- Other architectures, such as Intel's x86, are ***complex instruction set computers*** **(CISC)**

Principle 3: Smaller is Faster

H&H p301-303

Instructions: Addition & Subtraction

C Code	RISC-V assembly code
`a = b + c;`	`add a, b, c`
`a = b - c;`	`sub a, b, c`

- **`Add/sub:`** mnemonic indicates operation to perform
- **`b, c:`** source operands (on which the operation is performed)
- **`a:`** destination operand (to which the result is written)

More complex code is handled by multiple RISC-V instructions.

C Code	RISC-V assembly code
`a = b + c - d;`	`add t, b, c  # t = b + c` `sub a, t, d  # a = t - d`

Based on: "*Digital Design and Computer Architecture (RISC-V Edition)*" by Sarah Harris and David Harris (H&H),

H&H p303

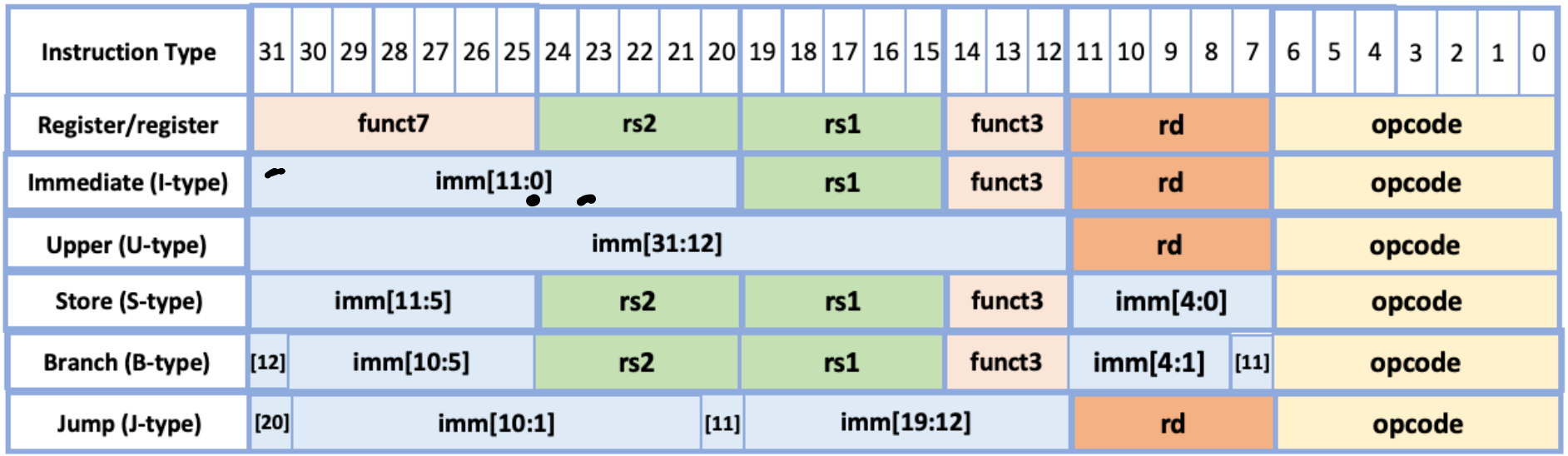
RISC-V Operands

- **Operand location:** physical location in computer
  - Registers
  - Memory
  - Constants (also called *immediates*)

- RISC-V has 32 32-bit registers
- Registers are faster than memory
- RISC-V called “32-bit architecture” because it operates on 32-bit data

H&H p304

32-bit RISC-V Instruction Types



- **opcode (7 bit):** partially specifies which of the 6 types of *instruction formats*
- **funct7 + funct3 (10 bit):** combined with **opcode**, these two fields describe what operation to perform
- **rs1 (5 bit):** specifies register containing first operand
- **rs2 (5 bit):** specifies second register operand
- **rd (5 bit):**: Destination register specifies register which will receive result of computation

RISC-V Registers

Name	Register Number	Usage
zero	x0	Constant value 0
ra	x1	Return address
sp	x2	Stack pointer
gp	x3	Global pointer
tp	x4	Thread pointer
t0-2	x5-7	Temporaries
s0/fp	x8	Saved register / Frame pointer
s1	x9	Saved register
a0-1	x10-11	Function arguments / return values
a2-7	x12-17	Function arguments
s2-11	x18-27	Saved registers
t3-6	x28-31	Temporaries

H&H p305

RISC-V operand from Registers

Name	Register Number	Usage
s0/fp	x8	Saved register / Frame pointer
s1	x9	Saved register
s2-11	x18-27	Saved registers

C Code

```
a = b + c;

a = b + 6;
```

RISC-V assembly code

```
# s0 = a, s1 = b, s2 = c
add s0, s1, s2

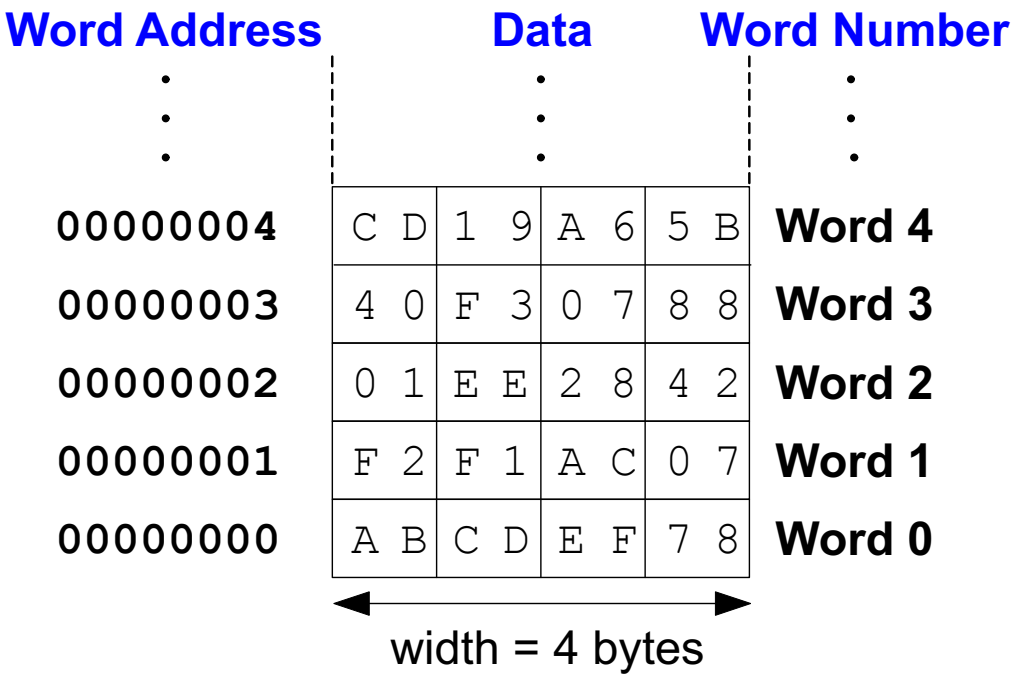
# s0 = a, s1 = b
addi s0, s1, 6
```



H&H p305

RISC-V operands from memory

- Each 32-bit data word has a unique address

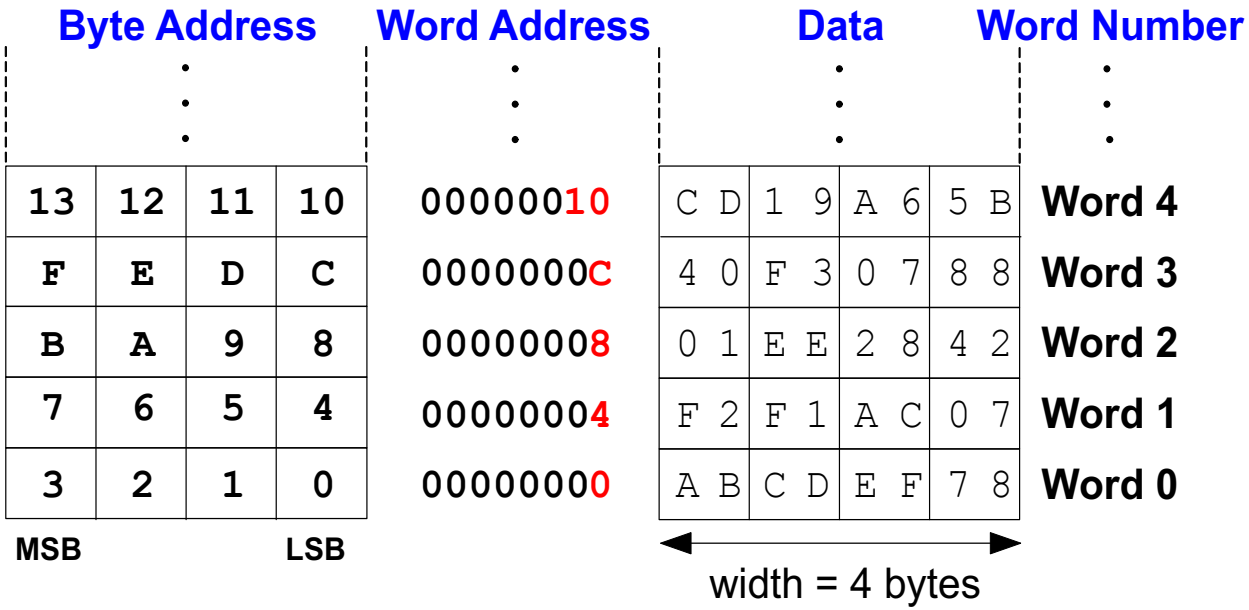


RISC-V uses **byte-addressable** memory (i.e. byte has a unique address), so each 32-bit word uses 4 byte addresses

H&H p307

RISC-V Byte-addressable Memory

- Each data byte has a unique address
- Load/store words or single bytes: load byte (lb) and store byte (sb)
- 32-bit word = 4 bytes, so word address **increments by 4**



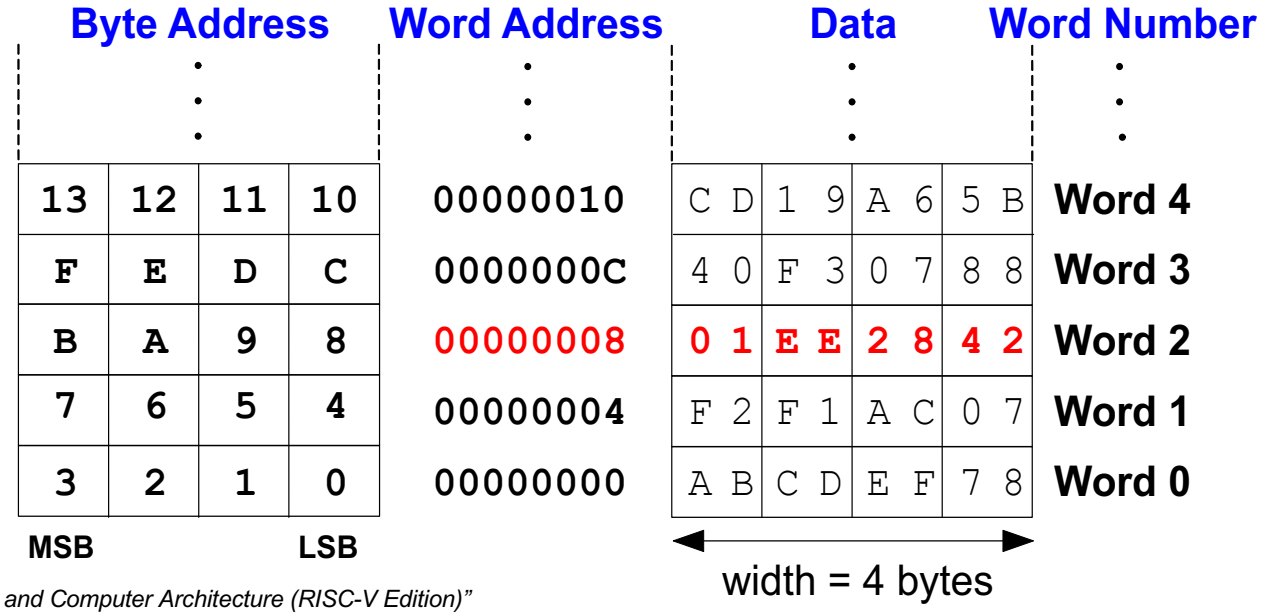
Based on: "*Digital Design and Computer Architecture (RISC-V Edition)*" by Sarah Harris and David Harris (H&H),

Reading Byte-Addressable Memory

- **Example:** Load a word of data at memory address 8 into s3.
- s3 holds the value 0x1EE2842 after load

RISC-V assembly code

```
lw s3, 8(zero)  # read word at address 8 into s3
```



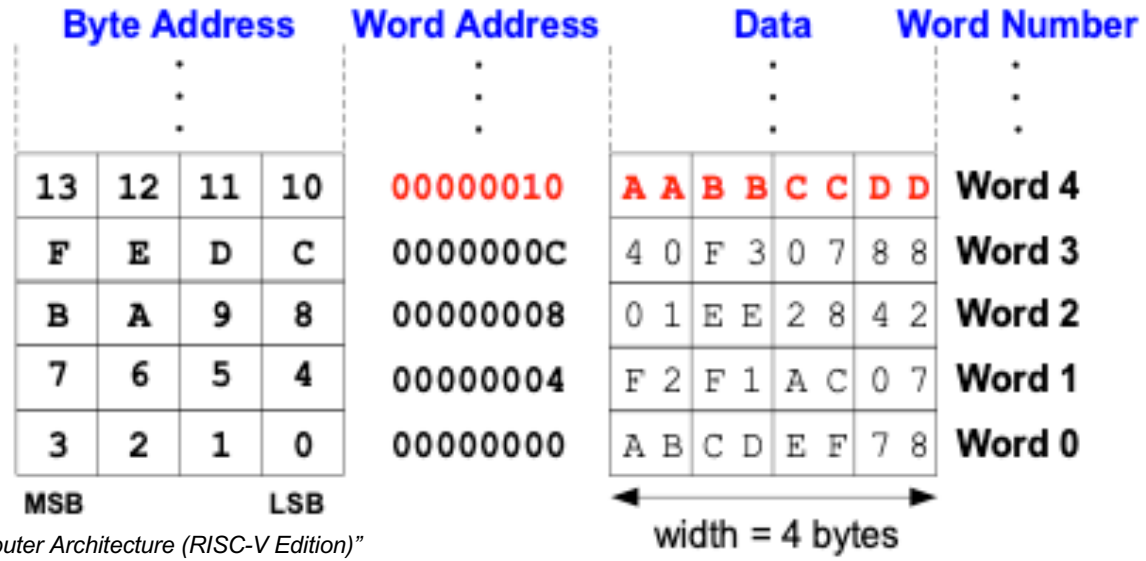
Based on: "*Digital Design and Computer Architecture (RISC-V Edition)*" by Sarah Harris and David Harris (H&H),

Writing Byte-Addressable Memory

- **Example:** store the value held in t7 into memory address 0x10 (16)
  - if t7 holds the value 0xAABBCCDD, then after the sw completes, word 4 (at address 0x10) in memory will contain that value

RISC-V assembly code

```
SW t7, 0x10(zero)  # write t7 into address 16
```



Based on: "*Digital Design and Computer Architecture (RISC-V Edition)*" by Sarah Harris and David Harris (H&H),

RISC-V: Operands from Constants

- 12-bit signed constants (immediates) using addi:

C Code

```
// int is a 32-bit signed word
int a = -372;
int b = a + 6;
```

RISC-V assembly code

```
# s0 = a, s1 = b
addi s0, zero, -372
addi s1, s0, 6
```

372 = 12'h174 = 12'b0001_0111_0100
-372 = 12'b1110_1000_1100 = 12'hE8B

- Form 32-bit constant using **sign extension**



Any immediate that needs **more than 12 bits** cannot use this method.

H&H p306

RISC-V: Operand with 32-bit Constants

- Use load upper immediate (lui) and addi
- lui: puts an immediate in the upper 20 bits of destination register and 0's in lower 12 bits

C Code

```
int a = 0xFEDC8765;
```

RISC-V assembly code

```
# s0 = a
lui  s0, 0xFEDC8
addi s0, s0, 0x765
```

Remember that addi **sign-extends** its 12-bit immediate constant



H&H p306

RISC-V: 32-bit Constants (bit 11 is 1)

- If **bit 11** of the constant is **1**, increment upper 20 bits by **1** in `lui`

C Code

```
int a = 0xFEDC8EAB;
```

Note: -341 = 0xEAB

RISC-V assembly code

```
# s0 = a
lui  s0, 0xFEDC9     # s0 = 0xFEDC9000
addi s0, s0, -341    # s0 = 0xFEDC9000 + 0xFFFFFEAB
                     #    = 0xFEDC8EAB
```

Based on: "*Digital Design and Computer Architecture (RISC-V Edition)*" by Sarah Harris and David Harris (H&H),

RISC-V: Psuedoinstruction

- Load immediate 32-bit word is tedious.
- Pseudoinstruction – Assembler program translate “Load Immediate” instruction “li” to two real RISC-V instructions: ”lui” and ”addi”

C Code

```
int a = 0xFEDC8EAB;
```

Note: -341 = 0xEAB

RISC-V pseudoinstructions

```
# s0 = a
li  s0, 0xFEDC8EAB
```

RISC-V real instructions

```
# s0 = a
lui  s0, 0xFEDC9
addi s0, s0, 0xEAB
```

- RISC-V has many pseudoinstructions (see later lectures)

RISC-V: Addressing Modes

How do we address the operands?

- Register Only
- Immediate
- Base Addressing
- PC-Relative

Register Only

- Operands found in registers
  - **Example:** add s0, t2, t3
  - **Example:** sub t6, s1, 0

Immediate

- 12-bit signed immediate used as an operand
  - **Example:** addi s4, t5, -73
  - **Example:** ori t3, t7, 0xFF

H&H p340

RISC-V: Base + Offset Addressing

Base Addressing

- Loads and Stores
- Address of operand is:

  base address + immediate

  – **Example:** lw s4, 72(zero)
    - address = 0 + 72

  – **Example:** sw t2, -25(t1)
    - address = t1 - 25

RISC-V: PC-relative Addressing

PC-Relative Addressing: branches and jal

Example:

Address		**Instruction**
0x354	L1:	addi s1, s1, 1
0x358		sub t0, t1, s7
…		…
0xEB0		bne s8, s9, L1

The label is (0xEB0-0x354) = 0xB5C (**2908**) instructions **before** bne

RISC-V: Instruction coding for Branch offset

Assembly

```
beq s8,  s9,  L1
(beq x24, x25, L1)
```

Relative offset = -2908

$imm_{12:0}$ = -2908 = `1 0 1 0 0 1 0 1 0 0 1 0 0`

bit number: `12 11 10 9 8 7 6 5 4 3 2 1 0`

Instruction Formats	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Branch	[12]	imm[10:5]						rs2					rs1					funct3			imm[4:1]				[11]	opcode						

Field Values

$imm_{12,10:5}$	rs2	rs1	funct3	$imm_{4:1,11}$	op
1100 101	24	25	1	0010 0	99
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

Machine Code

$imm_{12,10:5}$	rs2	rs1	funct3	$imm_{4:1,11}$	op
1100 101	11000	11001	001	0010 0	110 0011
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

`(0xCB8C9263)`

R-type Instructions: 3 register instructions

Instruction Formats	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Register/register	funct7							rs2					rs1					funct3			rd					opcode						

op	funct3	funct7	Type	Instruction	Description	Operation
0110011 (51)	000	0000000	R	`add rd, rs1, rs2`	add	`rd = rs1 + rs2`
0110011 (51)	000	0100000	R	`sub rd, rs1, rs2`	sub	`rd = rs1 - rs2`
0110011 (51)	001	0000000	R	`sll rd, rs1, rs2`	shift left logical	`rd = rs1 << rs2`$_{4:0}$
0110011 (51)	010	0000000	R	`slt rd, rs1, rs2`	set less than	`rd = (rs1 < rs2)`
0110011 (51)	011	0000000	R	`sltu rd, rs1, rs2`	set less than unsigned	`rd = (rs1 < rs2)`
0110011 (51)	100	0000000	R	`xor rd, rs1, rs2`	xor	`rd = rs1 ^ rs2`
0110011 (51)	101	0000000	R	`srl rd, rs1, rs2`	shift right logical	`rd = rs1 >>` `rs2`$_{4:0}$
0110011 (51)	101	0100000	R	`sra rd, rs1, rs2`	shift right arithmetic	`rd = rs1 >>>` `rs2`$_{4:0}$
0110011 (51)	110	0000000	R	`or rd, rs1, rs2`	or	`rd = rs1 | rs2`
0110011 (51)	111	0000000	R	`and rd, rs1, rs2`	and	`rd = rs1 & rs2`

I & S-type Instructions: All involve imm constants

Instruction Formats	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Immediate	imm[11:0]												rs1					funct3			rd					opcode						
Store	imm[11:5]							rs2					rs1					funct3			imm[4:0]					opcode						

op	funct3	funct7	Type	Instruction	Description	Operation
0000011 (3)	000	–	I	`lb rd, imm(rs1)`	load byte	rd = SignExt([Address]$_{7:0}$)
0000011 (3)	001	–	I	`lh rd, imm(rs1)`	load half	rd = SignExt([Address]$_{15:0}$)
0000011 (3)	010	–	I	`lw rd, imm(rs1)`	load word	rd = [Address]$_{31:0}$
0000011 (3)	100	–	I	`lbu rd, imm(rs1)`	load byte unsigned	rd = ZeroExt([Address]$_{7:0}$)
0000011 (3)	101	–	I	`lhu rd, imm(rs1)`	load half unsigned	rd = ZeroExt([Address]$_{15:0}$)
0010011 (19)	000	–	I	`addi rd, rs1, imm`	add immediate	rd = rs1 + SignExt(imm)
0010011 (19)	001	0000000*	I	`slli rd, rs1, uimm`	shift left logical immediate	rd = rs1 << uimm
0010011 (19)	010	–	I	`slti rd, rs1, imm`	set less than immediate	rd = (rs1 < SignExt(imm))
0010011 (19)	011	–	I	`sltiu rd, rs1, imm`	set less than imm. unsigned	rd = (rs1 < SignExt(imm))
0010011 (19)	100	–	I	`xori rd, rs1, imm`	xor immediate	rd = rs1 ^ SignExt(imm)
0010011 (19)	101	0000000*	I	`srli rd, rs1, uimm`	shift right logical immediate	rd = rs1 >> uimm
0010011 (19)	101	0100000*	I	`srai rd, rs1, uimm`	shift right arithmetic imm.	rd = rs1 >>> uimm
0010011 (19)	110	–	I	`ori rd, rs1, imm`	or immediate	rd = rs1 | SignExt(imm)
0010011 (19)	111	–	I	`andi rd, rs1, imm`	and immediate	rd = rs1 & SignExt(imm)
0100011 (35)	000	–	S	`sb rs2, imm(rs1)`	store byte	[Address]$_{7:0}$ = rs2$_{7:0}$
0100011 (35)	001	–	S	`sh rs2, imm(rs1)`	store half	[Address]$_{15:0}$ = rs2$_{15:0}$
0100011 (35)	010	–	S	`sw rs2, imm(rs1)`	store word	[Address]$_{31:0}$ = rs2

B-type Instructions: PC-relative Branches

Instruction Formats	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Branch	[12]	imm[10:5]						rs2					rs1					funct3			imm[4:1]				[11]	opcode						

op	funct3	funct7	Type	Instruction	Description	Operation
1100011 (99)	000	–	B	`beq rs1, rs2, label`	branch if =	`if (rs1 == rs2) PC = BTA`
1100011 (99)	001	–	B	`bne rs1, rs2, label`	branch if ≠	`if (rs1 ≠ rs2) PC = BTA`
1100011 (99)	100	–	B	`blt rs1, rs2, label`	branch if <	`if (rs1 < rs2) PC = BTA`
1100011 (99)	101	–	B	`bge rs1, rs2, label`	branch if ≥	`if (rs1 ≥ rs2) PC = BTA`
1100011 (99)	110	–	B	`bltu rs1, rs2, label`	branch if < unsigned	`if (rs1 < rs2) PC = BTA`
1100011 (99)	111	–	B	`bgeu rs1, rs2, label`	branch if ≥ unsigned	`if (rs1 ≥ rs2) PC = BTA`

H&H p311

U & I -type Instructions: Upper & Jump/Link



op	funct3	funct7	Type	Instruction	Description	Operation
0010111 (23)	–	–	U	`auipc rd, upimm`	add upper immediate to PC	`rd = {upimm, 12'b0} + PC`
0110111 (55)	–	–	U	`lui rd, upimm`	load upper immediate	`rd = {upimm, 12'b0}`
1100111 (103)	000	–	I	`jalr rd, rs1, imm`	jump and link register	`PC = rs1 + SignExt(imm), rd = PC + 4`
1101111 (111)	–	–	J	`jal rd, label`	jump and link	`PC = JTA, rd = PC + 4`

- We will discuss auipc, jalr and jal instructions in another lecture

RISC-V Arithmetic instructions

Mnemonic	Instruction	Type	Description
ADD rd, rs1, rs2	Add	R	rd ← rs1 + rs2
SUB rd, rs1, rs2	Subtract	R	rd ← rs1 - rs2
ADDI rd, rs1, imm12	Add immediate	I	rd ← rs1 + imm12
SLT rd, rs1, rs2	Set less than	R	rd ← rs1 < rs2 ? 1 : 0
SLTI rd, rs1, imm12	Set less than immediate	I	rd ← rs1 < imm12 ? 1 : 0
SLTU rd, rs1, rs2	Set less than unsigned	R	rd ← rs1 < rs2 ? 1 : 0
SLTIU rd, rs1, imm12	Set less than immediate unsigned	I	rd ← rs1 < imm12 ? 1 : 0
LUI rd, imm20	Load upper immediate	U	rd ← imm20 << 12
AUIP rd, imm20	Add upper immediate to PC	U	rd ← PC + imm20 << 12

RISC-V Logic instructions

Mnemonic	Instruction	Type	Description
AND rd, rs1, rs2	AND	R	rd ← rs1 & rs2
OR rd, rs1, rs2	OR	R	rd ← rs1 | rs2
XOR rd, rs1, rs2	XOR	R	rd ← rs1 ^ rs2
ANDI rd, rs1, imm12	AND immediate	I	rd ← rs1 & imm12
ORI rd, rs1, imm12	OR immediate	I	rd ← rs1 | imm12
XORI rd, rs1, imm12	XOR immediate	I	rd ← rs1 ^ imm12
SLL rd, rs1, rs2	Shift left logical	R	rd ← rs1 << rs2
SRL rd, rs1, rs2	Shift right logical	R	rd ← rs1 >> rs2
SRA rd, rs1, rs2	Shift right arithmetic	R	rd ← rs1 >> rs2
SLLI rd, rs1, shamt	Shift left logical immediate	I	rd ← rs1 << shamt
SRLI rd, rs1, shamt	Shift right logical imm.	I	rd ← rs1 >> shamt
SRAI rd, rs1, shamt	Shift right arithmetic immediate	I	rd ← rs1 >> shamt

RISC-V Load/Store instructions

Mnemonic	Instruction	Type	Description
`LW rd, imm12(rs1)`	Load word	I	`rd ← mem[rs1 + imm12]`
`LH rd, imm12(rs1)`	Load halfword	I	`rd ← mem[rs1 + imm12]`
`LB rd, imm12(rs1)`	Load byte	I	`rd ← mem[rs1 + imm12]`
`LWU rd, imm12(rs1)`	Load word unsigned	I	`rd ← mem[rs1 + imm12]`
`LHU rd, imm12(rs1)`	Load halfword unsigned	I	`rd ← mem[rs1 + imm12]`
`LBU rd, imm12(rs1)`	Load byte unsigned	I	`rd ← mem[rs1 + imm12]`
`SW rs2, imm12(rs1)`	Store word	S	`rs2(31:0) → mem[rs1 + imm12]`
`SH rs2, imm12(rs1)`	Store halfword	S	`rs2(15:0) → mem[rs1 + imm12]`
`SB rs2, imm12(rs1)`	Store byte	S	`rs2(7:0) → em[rs1 + imm12]`

RISC-V Branch & Jump instructions

Mnemonic	Instruction	Type	Description
`BEQ rs1, rs2, imm12`	Branch equal	SB	`if rs1 == rs2` `  pc ← pc + imm12`
`BNE rs1, rs2, imm12`	Branch not equal	SB	`if rs1 != rs2` `  pc ← pc + imm12`
`BGE rs1, rs2, imm12`	Branch greater than or equal	SB	`if rs1 >= rs2` `  pc ← pc + imm12`
`BGEU rs1, rs2, imm12`	Branch greater than or equal unsigned	SB	`if rs1 >= rs2` `  pc ← pc + imm12`
`BLT rs1, rs2, imm12`	Branch less than	SB	`if rs1 < rs2` `  pc ← pc + imm12`
`BLTU rs1, rs2, imm12`	Branch less than unsigned	SB	`if rs1 < rs2` `  pc ← pc + imm12 << 1`
`JAL rd, imm20`	Jump and link	UJ	`rd ← pc + 4` `pc ← pc + imm20`
`JALR rd, imm12(rs1)`	Jump and link register	I	`rd ← pc + 4` `pc ← rs1 + imm12`

RISC-V Psuedoinstructions

Mnemonic	Instruction	Base instruction(s)
`LI rd, imm12`	Load immediate (near)	`ADDI rd, zero, imm12`
`LI rd, imm`	Load immediate (far)	`LUI rd, imm[31:12]` `ADDI rd, rd, imm[11:0]`
`LA rd, sym`	Load address (far)	`AUIPC rd, sym[31:12]` `ADDI rd, rd, sym[11:0]`
`MV rd, rs`	Copy register	`ADDI rd, rs, 0`
`NOT rd, rs`	One's complement	`XORI rd, rs, -1`
`NEG rd, rs`	Two's complement	`SUB rd, zero, rs`
`BGT rs1, rs2, offset`	Branch if rs1 > rs2	`BLT rs2, rs1, offset`
`BLE rs1, rs2, offset`	Branch if rs1 ≤ rs2	`BGE rs2, rs1, offset`
`BGTU rs1, rs2, offset`	Branch if rs1 > rs2 (unsigned)	`BLTU rs2, rs1, offset`
`BLEU rs1, rs2, offset`	Branch if rs1 ≤ rs2 (unsigned)	`BGEU rs2, rs1, offset`

Mnemonic	Instruction	Base instruction(s)
`BEQZ rs1, offset`	Branch if rs1 = 0	`BEQ rs1, zero, offset`
`BNEZ rs1, offset`	Branch if rs1 ≠ 0	`BNE rs1, zero, offset`
`BGEZ rs1, offset`	Branch if rs1 ≥ 0	`BGE rs1, zero, offset`
`BLEZ rs1, offset`	Branch if rs1 ≤ 0	`BGE zero, rs1, offset`
`BGTZ rs1, offset`	Branch if rs1 > 0	`BLT zero, rs1, offset`
`J offset`	Unconditional jump	`JAL zero, offset`
`CALL offset12`	Call subroutine (near)	`JALR ra, ra, offset12`
`CALL offset`	Call subroutine (far)	`AUIPC ra, offset[31:12]` `JALR ra, ra, offset[11:0]`
`RET`	Return from subroutine	`JALR zero, 0(ra)`
`NOP`	No operation	`ADDI zero, zero, 0`